PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date	25-09-2026
Team ID	
Project Name	Medical Diagnosis Center by Service Portal (ServiceNow)
Maximum Marks	4 Marks

Technical Architecture

The architecture below presents the proposed components and data flow for the Medical Diagnosis Center ServiceNow Service Portal. It identifies user interfaces, application logic, platform data storage, report attachments, infrastructure, and optional external interfaces, following the architecture guidelines in the provided template.

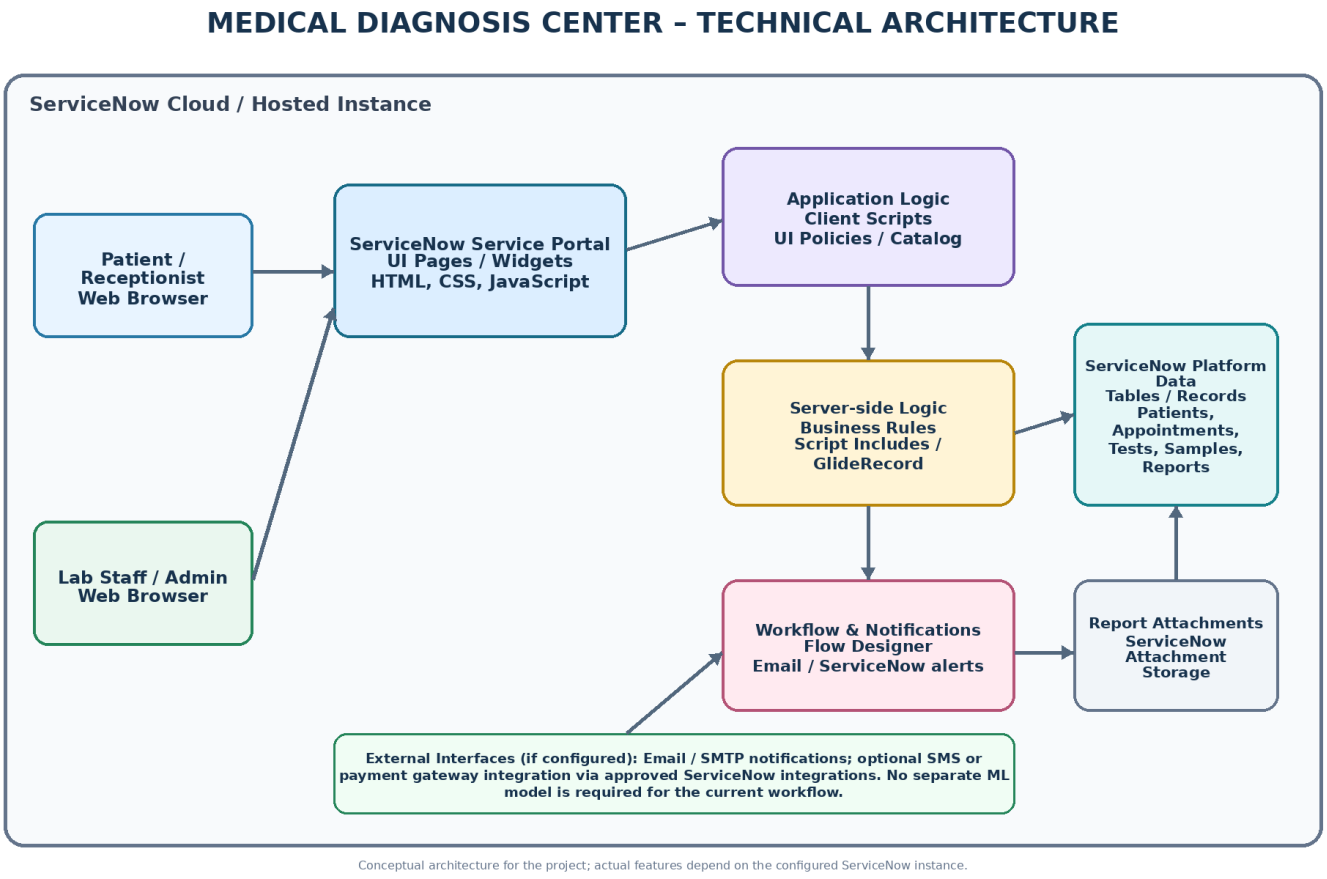


Figure 1. Conceptual technical architecture of the Medical Diagnosis Center Service Portal.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Patient, receptionist, lab staff, and administrator interact with the portal through browser-based pages and widgets.	ServiceNow Service Portal, HTML, CSS, JavaScript
2	Application Logic-1	Patient registration, login access, test catalog display, and appointment booking validation.	ServiceNow UI Policies, Client Scripts, Catalog / Portal logic
3	Application Logic-2	Server-side processing for bookings,	Business Rules, Script Includes,

		sample updates, report association, and record validation.	GlideRecord (JavaScript)
4	Application Logic-3	Workflow automation for appointment states, sample processing, report-ready alerts, and administrative tasks.	Flow Designer, ServiceNow notifications
5	Database	Structured records for patient profiles, diagnostic tests, appointments, sample status, and report metadata.	ServiceNow platform tables / records
6	Cloud Database	Platform-managed data storage hosted as part of the ServiceNow instance; no separate cloud database is specified in the project.	ServiceNow instance data storage
7	File Storage	Storage of uploaded diagnostic report files and related attachments, subject to configured access controls.	ServiceNow Attachment storage
8	External API-1	Email notifications for booking confirmation and report/status updates, where configured.	ServiceNow Email / SMTP configuration
9	External API-2	Optional SMS or payment gateway integration if enabled and configured; not assumed as already implemented.	Approved ServiceNow integration / REST API (optional)
10	Machine Learning Model	The described project focuses on workflow and record management; a separate ML model is not required for the current scope.	Not applicable to current project scope
11	Infrastructure (Server / Cloud)	Portal and server-side application run on the hosted ServiceNow platform; users access it through a web browser.	ServiceNow cloud instance; browser-based client

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The project uses the ServiceNow platform and its portal framework. HTML, CSS, and JavaScript are used for portal UI customization; no additional open-source framework is specified.	ServiceNow Service Portal, HTML, CSS, JavaScript
2	Security Implementations	Access should be controlled through authentication, roles, ACLs, and restricted access to patient records and reports. Sensitive report attachments should only be available to authorized users.	ServiceNow roles, Access Control Lists (ACLs), platform authentication
3	Scalable Architecture	The application uses a platform-based, modular structure separating portal UI, business logic, workflows, and data records. Capacity is subject to the ServiceNow instance and configuration.	ServiceNow platform architecture; modular portal components
4	Availability	Access is provided through the hosted ServiceNow instance, subject to platform service availability and scheduled maintenance.	ServiceNow cloud hosting
5	Performance	Use efficient queries, avoid unnecessary client/server calls, and keep portal forms and dashboards focused on required records. Actual throughput depends on instance configuration and load.	GlideRecord query practices, optimized scripts and portal components